



CentraleSupélec

Reinforcement Learning

Project Report – Highway-v0

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Abstract

This report presents my individual contributions to the group RL project on the **highway-v0** environment. My work covers two interconnected tasks: (i) the design and execution of a hyperparameter search pipeline for a custom DQN using Optuna, and (ii) the implementation, training, and behavioral analysis of a Double DQN agent as part of the group’s ablation study. The central question I investigated is: *does systematic hyperparameter optimization close the performance gap between a vanilla DQN and its algorithmic extensions, or do architectural improvements provide independent gains?*

Repository: <https://github.com/Nicolas2412/rl-highway>

Group members: Nicolas Charrondière, Adam Douss, Paul Guimbert, **Sarah Lamik**

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1 Introduction

1.1 Notation

Symbol	Definition
$s_t \in \mathcal{S}$	State at time t
$a_t \in \mathcal{A}$	Action at time t
$r_t \in \mathbb{R}$	Reward received at time t
$\gamma \in (0, 1)$	Discount factor
$Q(s, a; \theta)$	Action-value function parametrized by θ
θ^-	Parameters of the target network
α	Learning rate
ε	Exploration rate (epsilon-greedy)
$ \mathcal{B} $	Replay buffer capacity

Table 1: Table of notations.

1.2 Environment and observation space

The `highway-v0` environment simulates a multi-lane highway in which the controlled agent must navigate among autonomously driven vehicles. The observation space is a *Kinematics* matrix of shape $(V \times F)$, where $V = 5$ is the number of observed vehicles (including the ego-vehicle) and $F = 5$ denotes the features $(x, y, v_x, v_y, \text{presence})$, all normalized to $[0, 1]$. The action space is `DiscreteMetaAction` with five actions: `LANE_LEFT`, `IDLE`, `LANE_RIGHT`, `FASTER`, `SLOWER`.

1.3 Reward function

The reward at each step is defined as:

$$r_t = w_v \cdot \mathbf{1}[v_t \in v_{\text{range}}] + w_r \cdot \frac{\ell}{\ell_{\max}} + w_c \cdot \mathbf{1}[\text{collision}] + w_{\Delta\ell} \cdot \mathbf{1}[\text{lane change}] \quad (1)$$

with the shared configuration:

$$w_v = 0.7, \quad w_r = 0.0, \quad w_c = -1.5, \quad w_{\Delta\ell} = -0.02, \quad v_{\text{range}} = [22, 30] \text{ m/s}, \quad \text{normalize_reward} = \text{True}.$$

This reward structure incentivizes high-speed driving while strongly penalizing collisions. The asymmetry between the speed reward (+0.7 per step) and the collision penalty (−1.5) is deliberately sharp: a single collision cancels approximately two full steps of maximum speed reward, creating strong pressure toward conservative collision avoidance. The lane-change penalty (−0.02 per step) is small but non-negligible over a long episode, marginally discouraging unnecessary lateral movements.

2 Core Task

2.1 Custom DQN: shared implementation

All group members implemented an independent version of the DQN agent as a first step toward familiarizing themselves with the environment. My personal implementation follows the standard DQN algorithm: a replay buffer of capacity $|\mathcal{B}|$, an online network $Q(\cdot; \theta)$ and a target

network $Q(\cdot; \theta^-)$ updated every τ steps, and an ε -greedy exploration schedule decaying linearly from $\varepsilon_{\text{start}} = 1.0$ to $\varepsilon_{\text{end}} = 0.01$ over T_ε steps. The Bellman target is:

$$y_t = r_t + \gamma \max_{a'} Q(s_{t+1}, a'; \theta^-) \quad (2)$$

and the loss minimized at each gradient step is the mean squared error:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{B}} \left[(y_t - Q(s, a; \theta))^2 \right]. \quad (3)$$

The global architecture (base agent class, training loop, evaluation harness) was defined collaboratively. I used this shared scaffold to implement both the fine-tuning pipeline and the Double DQN variant described in later sections.

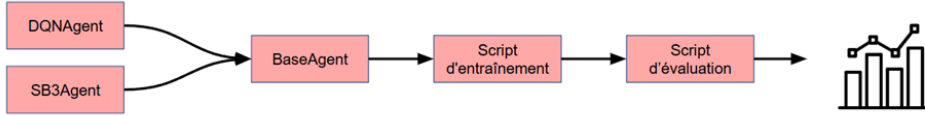


Figure 1: Project pipeline and shared agent architecture.

2.2 Fine-tuning the vanilla DQN with Optuna

This section describes work I designed and implemented independently.

2.2.1 Methodological framework

Sampler. Hyperparameter search was performed with Optuna using the *Tree-structured Parzen Estimator* (TPE) sampler. TPE is preferred over grid search because it models the distribution of high-performing and low-performing configurations separately, then samples preferentially from the promising region. This Bayesian strategy is critically more sample-efficient than grid search in the RL setting, where a single trial requires several thousand environment steps.

Pruner and evaluation protocol. A MedianPruner with a warm-up of 5 trials was used: a trial is pruned mid-training if its intermediate objective falls below the median of all completed trials at the same training step. The 5-trial warm-up ensures sufficient reference points before pruning decisions are made. Every 1 000 steps, the trial’s running performance is evaluated as the mean reward over the last 20 episodes. Each trial runs for a maximum of 10 000 steps, sufficient to discriminate underperforming configurations without committing to a full training run. If no episode was completed in the last 20 logged steps, the objective is set to -999 to strongly penalize collapsed policies. For reproducibility, all trials are run with a fixed seed.

Search space. The search space is summarized in Table 2. The learning rate is sampled on a log-uniform scale in $[10^{-4}, 10^{-3}]$, as learning-rate effects are multiplicative. The discount factor γ spans $[0.80, 0.99]$: the lower bound tests whether a shorter-horizon agent performs better in this episodic, crash-heavy environment. Batch size, hidden size, and buffer capacity are categorical, covering the standard powers-of-two used in deep RL. The target update frequency $\tau \in [25, 200]$ controls the stability-tracking trade-off of the target network. The ε -decay horizon $T_\varepsilon \in [50\,000, 150\,000]$ governs how long the agent explores before committing to its learned policy.

Hyperparameter	Search space	Type
α (learning rate)	$[10^{-4}, 10^{-3}]$	Log-uniform
γ (discount)	$[0.80, 0.99]$	Uniform
batch_size	$\{32, 64, 128\}$	Categorical
hidden_size	$\{128, 256, 512\}$	Categorical
n_{layers}	$\{1, 2, 3\}$	Integer
τ (target update freq.)	$[25, 200]$	Integer
T_ε (eps. decay steps)	$[50\,000, 150\,000]$	Integer
$ \mathcal{B} $ (buffer capacity)	$\{10\text{k}, 20\text{k}, 30\text{k}\}$	Categorical

Table 2: Hyperparameter search space for the Optuna fine-tuning study.

2.2.2 Results of the hyperparameter search

Optimization history and convergence. Figure 2 presents the optimization history (left) and the fANOVA-based hyperparameter importance scores (right). Three out of ten trials were pruned, confirming that the pruner successfully eliminated low-performing configurations early. The best configuration (trial #0) achieved a mean reward of 13.31 over the last 20 episodes of a 10 000-step trial, and the cumulative best remained stable thereafter: an early convergence that reflects the small trial budget rather than an exhaustive coverage of the search space.

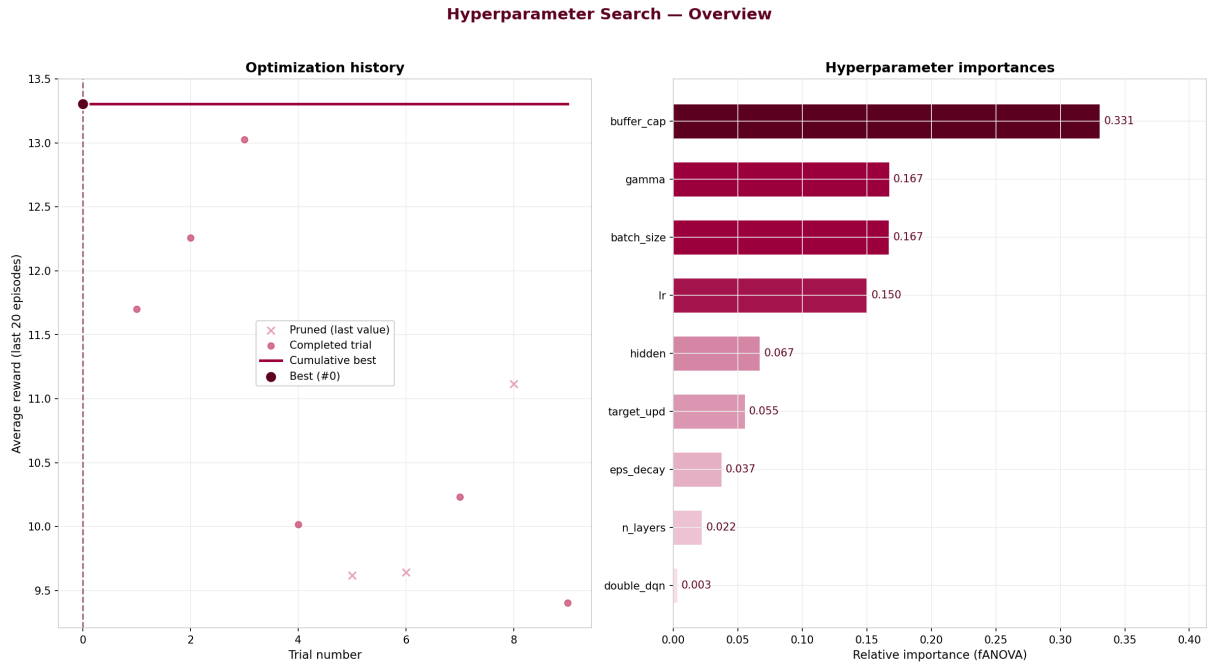


Figure 2: Left: Optimization history. Crosses denote pruned trials; filled dots are completed trials. The dark horizontal line tracks the cumulative best (trial #0, reward = 13.31). Right: Hyperparameter importances computed via fANOVA. Buffer capacity dominates with a relative importance of 0.331.

Hyperparameter importance. The fANOVA decomposition reveals a clear hierarchy. Buffer capacity $|\mathcal{B}|$ is by far the most influential parameter (importance = 0.331), followed by γ and batch size (both ≈ 0.167), then the learning rate (0.150). Architectural choices (hidden size, n_{layers}) have markedly lower importance (0.067 and 0.022 respectively). Notably, ε -decay and target update frequency, often highlighted as critical in the DQN literature, contribute only

0.037 and 0.055, suggesting that in the **highway-v0** setting the quality of stored experience matters more than the exploration schedule.

The dominance of $|\mathcal{B}|$ carries a practical message: in short-episode, high-collision environments, a larger buffer retains rare but highly informative transitions (near-miss avoidances, successful overtakings) for longer, which stabilizes gradient estimates more than architectural depth or target network frequency.

Continuous parameters. Figure 3 shows reward as a function of each continuous hyperparameter. The learning rate trend is weakly positive across the explored range $[10^{-4}, 7 \times 10^{-4}]$, consistent with the best trial’s $\alpha = 1.43 \times 10^{-4}$. More strikingly, the relationship with γ exhibits a negative slope: rewards are highest around $\gamma \approx 0.83$ and decline toward $\gamma \approx 0.99$. This counter-intuitive finding suggests that a shorter planning horizon helps in this environment: a possible explanation is that a low γ implicitly down-weights the long-term consequences of risky high-speed decisions, making the agent more responsive to immediate collision signals. For T_ϵ and τ , the data points are too sparse to draw firm conclusions.

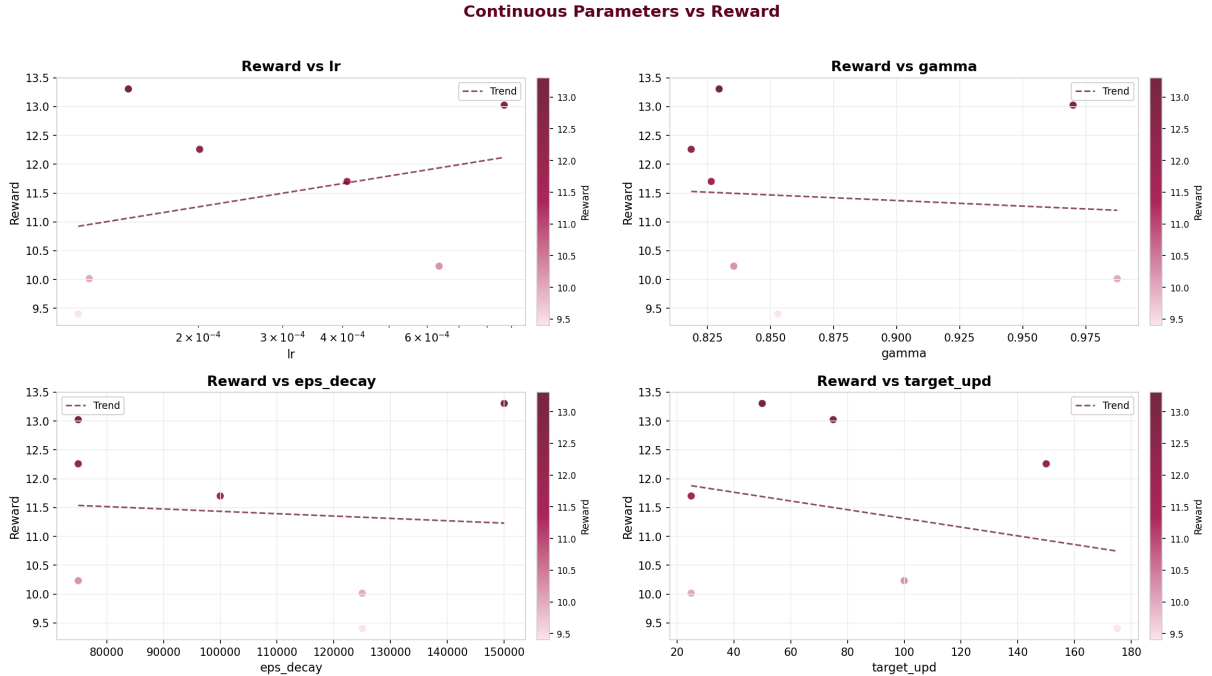


Figure 3: Reward as a function of each continuous hyperparameter (completed trials only). Trend lines are linear fits for visual guidance; the low trial count ($n = 7$ completed) limits statistical significance.

Categorical parameters. Figure 4 displays reward distributions by categorical value. For batch size, 128 is clearly dominant (median reward ≈ 13.1), while 32 performs worst, consistent with the high gradient variance expected at small batch sizes in RL. For buffer capacity, $|\mathcal{B}| = 30\,000$ yields the highest rewards (median ≈ 12.9), while 20,000 performs worse than 10,000 (median ≈ 9.3). This non-monotonic result is most likely a confound of the small sample size (7 trials) rather than a causal effect and should not be over-interpreted. For hidden size, 256 and 512 neurons perform comparably and both outperform 128.

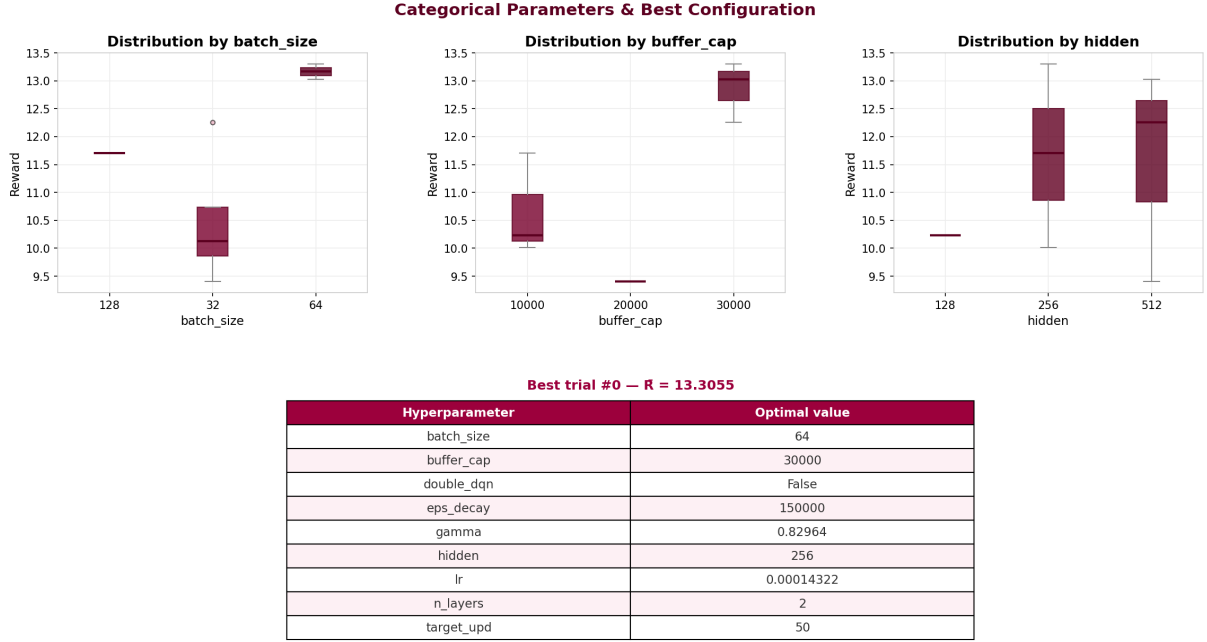


Figure 4: Reward distributions by categorical hyperparameter (box plots over completed trials). Bottom table: optimal configuration found by Optuna (trial #0, reward = 13.31).

Optimal configuration. The best trial (#0) yielded the configuration in Table 3.

Hyperparameter	Optimal value
α	1.43×10^{-4}
γ	0.830
batch_size	64
hidden_size	256
n_{layers}	2
τ (target update freq.)	50
T_{ϵ} (eps. decay steps)	150 000
$ \mathcal{B} $ (buffer capacity)	30 000

Table 3: Optimal hyperparameter configuration (trial #0, reward = 13.31).

Note on division of labor: the fine-tuned model was subsequently trained to convergence (200 000 steps) by another group member. I personally designed the Optuna pipeline, ran the search, and conducted all comparative evaluations of the fine-tuned vs. default DQN reported below.

2.2.3 Impact of fine-tuning: quantitative evaluation

Figures 5–7 compare the fine-tuned and default DQN evaluated over 3×50 episodes.

Summary metrics (Figure 5). The fine-tuned model achieves a mean reward of ≈ 21 versus ≈ 10 for the untuned model, a $\approx 110\%$ **relative improvement**. The most dramatic effect is on survival rate: the untuned model survives only $\approx 4\%$ of episodes, essentially indistinguishable from the random baseline ($\approx 7\%$), while the fine-tuned model survives $\approx 58\%$ of episodes, a 54-percentage-point gap. Mean survival duration grows from ≈ 13 to ≈ 24 steps (+85%) and mean speed increases from ≈ 25.0 to ≈ 26.4 m/s. The error bars on the fine-tuned model are

small relative to the gap between models, confirming that these differences are not an artefact of seed variance.

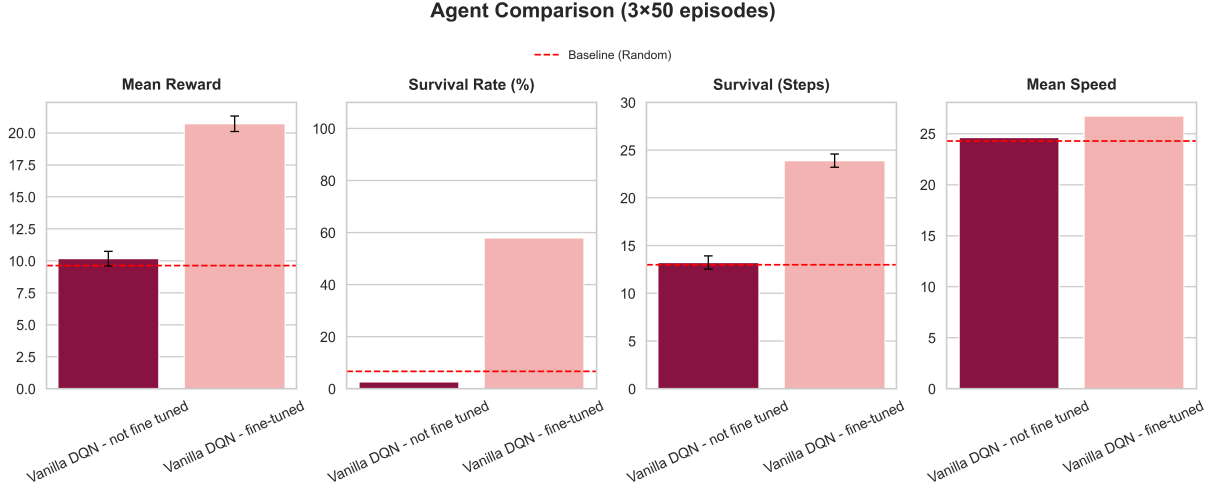


Figure 5: Agent comparison (fine-tuned vs. default DQN) over 3×50 evaluation episodes. Four metrics are reported: mean reward, survival rate (%), survival duration (steps), and mean speed (m/s). The red dashed line indicates the random baseline. Error bars represent the standard error across the 3 evaluation seeds.

Reward distribution (Figure 6). The violin plot reveals a qualitative difference in policy structure beyond the mean shift. The untuned DQN produces a unimodal distribution concentrated between 5 and 27, with mean ≈ 10.3 , corresponding to a single behavioral regime dominated by early collision. The fine-tuned DQN produces a **bimodal distribution**: a lower mode around 10–14 (short episodes ending in collision) and a dominant upper mode around 22–29 (long, high-speed trajectories). This bimodality is a structural finding: the fine-tuned agent has not simply improved on average but has acquired two distinct behavioral modes, a residual failure mode and a competent cruising mode that it exhibits in the majority of episodes.

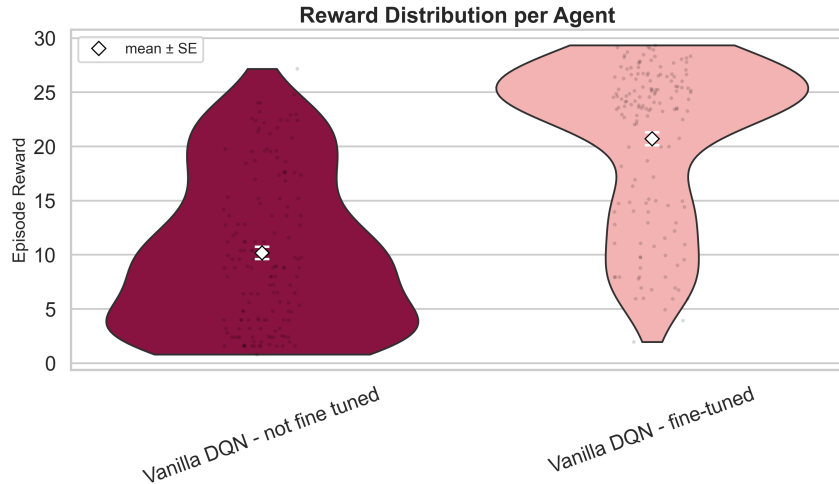


Figure 6: Episode reward distributions (violin plot with individual episode points and mean $\pm$ SE markers). The untuned DQN is unimodal (mean ≈ 10.3). The fine-tuned DQN is bimodal with a dominant upper mode at 22–29, reflecting two distinct behavioral regimes.

Speed distribution (Figure 7, top). The per-step speed distributions expose a fundamental difference in the effective action repertoire of the two policies. The untuned DQN produces an

almost degenerate distribution: a near-Dirac spike at $v \approx 25$ m/s (density ≈ 1.0 , KDE half-width ≈ 1 m/s) with a residual bump near 30 m/s. This near-zero variance indicates that the untuned agent has collapsed to a single-speed policy: it maintains a fixed cruising speed within the reward range but does *not* modulate speed in response to traffic. The FASTER and SLOWER actions are effectively unused.

The fine-tuned DQN’s speed distribution is qualitatively different: broadly bimodal with peaks at ≈ 20 m/s and ≈ 29 m/s, mean = 26.4 m/s, and non-negligible density below 22 m/s. The presence of mass below the reward speed range demonstrates that the fine-tuned agent has internalized that *briefly sacrificing speed reward to avoid a collision is advantageous*, a non-trivial credit assignment that the untuned model failed to acquire.

Survival curve (Figure 7, bottom). The Kaplan-Meier-style survival curve $\hat{P}(\text{episode length} > t)$ makes the gap maximally clear. The untuned DQN overlaps almost exactly with the random baseline: at step 20, only $\approx 22\%$ of its episodes remain active. The fine-tuned DQN diverges sharply from step 3 onward and maintains $\approx 70\%$ survival through step 20.

A notable feature is the **sharp drop between steps 28 and 30**, from $\approx 62\%$ to 0%. This concentrated late-episode collapse is a genuine and distinct failure mode, consistent with the behavioral analysis in Section 3.6: the fine-tuned agent’s strategy of prioritizing speed over spatial reorganization eventually breaks down in high-density situations that develop over longer episodes.

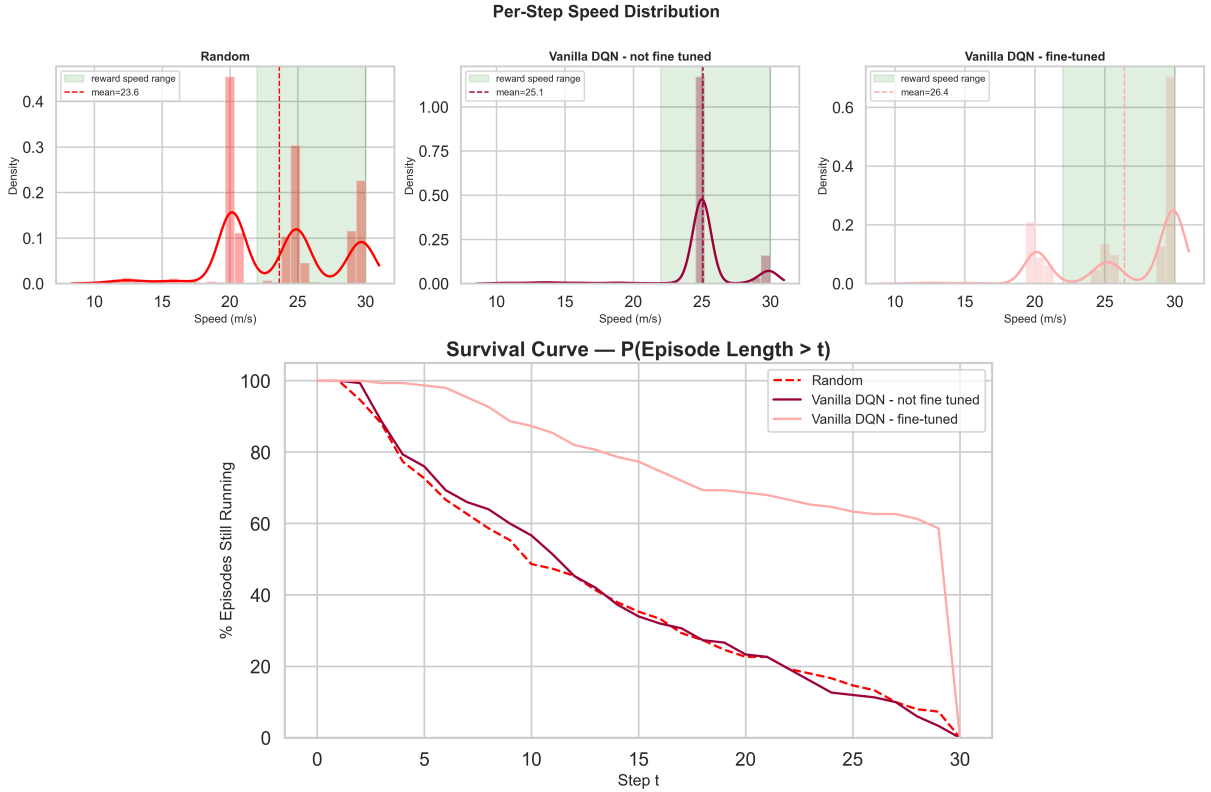


Figure 7: Top: per-step speed distributions for the random agent, the untuned DQN, and the fine-tuned DQN. The untuned model collapses to a near-Dirac at 25 m/s; the fine-tuned model exhibits a broad bimodal distribution (peaks at ≈ 20 and ≈ 29 m/s, mean = 26.4 m/s). Bottom: survival curves $\hat{P}(\text{episode length} > t)$. The untuned DQN tracks the random baseline; the fine-tuned DQN maintains $\approx 70\%$ survival at step 20 before collapsing sharply between steps 28–30.

2.2.4 Limitations of the fine-tuning study

Trial budget. Only 10 trials (7 completed, 3 pruned) were run. The fANOVA importance estimates are unreliable with such a small sample; a budget of 50–100 trials would be the minimum for robust conclusions.

Coverage of continuous variables. Figure 3 reveals large unexplored regions: no trial explored $\alpha > 7 \times 10^{-4}$ or $\tau > 185$. Identified trends should be treated as preliminary hypotheses.

Single-seed tuning. The entire Optuna study used a fixed seed. A multi-seed objective (mean reward over 3 seeds per trial) would yield a more robust configuration at the cost of $3\times$ the compute.

Short trials vs. full training. Each trial runs 10 000 steps; full training runs 200 000 steps. There is no guarantee that the configuration optimal at step 10 000 is also optimal at convergence, a well-documented issue in hyperparameter optimization for RL. The relatively modest final performance of the fine-tuned model in the full ablation comparison (Section 3.5) is consistent with this concern.

3 Extension Task: Ablation Study

3.1 Motivation and design

I proposed the ablation study as the group’s extension task. Beyond its practical advantage of distributing work cleanly among group members, it addresses a precise question: *what is the independent contribution of each algorithmic component added on top of the vanilla DQN?* The five models in the study are presented in Table 4.

Model	Tuned?	Algorithmic addition
SB3 DQN	No	Stable-Baselines3 baseline
DQN (fine-tuned)	Yes (Optuna)	Hyperparameter optimization
Double DQN	No	Decoupled action selection / evaluation
DQN + PER	No	Prioritized experience replay
DDQN + PER	No	Both

Table 4: Ablation study design. The DDQN and PER variants are intentionally left untuned to isolate the independent contribution of each algorithmic component.

The deliberate asymmetry of tuning only the vanilla DQN answers the question: *can a well-tuned simple model match or outperform untuned, more complex models?*

3.2 Double DQN: theoretical motivation

I was responsible for implementing, training, and analyzing the Double DQN.

The standard DQN update (Eq. 2) uses the target network both to *select* the greedy action and to *evaluate* its value:

$$y_t^{\text{DQN}} = r_t + \gamma \underbrace{\max_{a'} Q(s_{t+1}, a'; \theta^-)}_{\text{selection and evaluation by } \theta^-} \quad . \quad (4)$$

Since the max operator applied to noisy estimates is positively biased ($\mathbb{E}[\max_i X_i] \geq \max_i \mathbb{E}[X_i]$), this leads to systematically overestimated Q-values, a phenomenon known as *maximization bias*.

Double DQN decouples selection and evaluation:

$$y_t^{\text{DDQN}} = r_t + \gamma Q \left(s_{t+1}, \underbrace{\arg \max_{a'} Q(s_{t+1}, a'; \theta)}_{\text{selection by online net}}; \underbrace{\theta^-}_{\text{evaluation by target net}} \right). \quad (5)$$

By using the online network to select and the target network to evaluate, the overestimation bias is significantly reduced. In a collision-penalized environment like `highway-v0`, this should translate into fewer collisions: the agent is less prone to overvaluing risky actions whose Q-values are inflated by the max operator.

3.3 Experimental setup

To ensure a clean ablation, all models except the fine-tuned DQN share the same hyperparameters (Table 5). PER-specific parameters ($\alpha_{\text{PER}} = 0.6$, $\beta_{\text{PER}} = 0.4$) are set to their canonical values to isolate each algorithmic contribution.

Hyperparameter	Value
α (learning rate)	5×10^{-4}
γ	0.9
batch_size	32
$ \mathcal{B} $ (buffer capacity)	15 000
total_timesteps	200 000
hidden_dims	[256, 256]
$\varepsilon_{\text{start}}$	1.0
ε_{end}	0.01
T_ε (eps. decay steps)	100 000
train_frequency	1
τ (target update freq.)	50
learning_starts	200
seed	42

Table 5: Shared hyperparameters for all untuned ablation models.

3.4 Training dynamics of the Double DQN

Figure 8 shows the training curves of the Double DQN over 200 000 steps ($\approx 17\,800$ episodes).

Reward. The sliding-window mean reward (window = 906 episodes) increases monotonically from ≈ 10 at episode 0 to ≈ 23 at episode 17 800. The high raw variance is expected: with $\varepsilon > 0.01$ throughout training (the decay reaches ε_{end} at step 100 000, i.e., around episode 8 700), random actions continue to cause collisions and inflate the per-episode variance.

Loss. The TD loss peaks sharply in the first $\approx 25\,000$ steps then decays toward ≈ 1 by step 100 000, consistent with the Q-function converging as the replay buffer fills with higher-quality transitions. Residual fluctuations beyond step 100 000 reflect the ongoing non-stationarity of the target network updates.

Episode length. The sliding mean episode length mirrors the reward curve, growing from ≈ 17 to ≈ 26 steps, providing a complementary behavioral signal that confirms improved collision avoidance.

Early stopping. Both the reward and episode length plateau near step 150 000. The last 50 000 training steps provide marginal gains; an early-stopping criterion (e.g., no improvement in sliding mean reward over 5 000 steps) would have been a worthwhile addition.

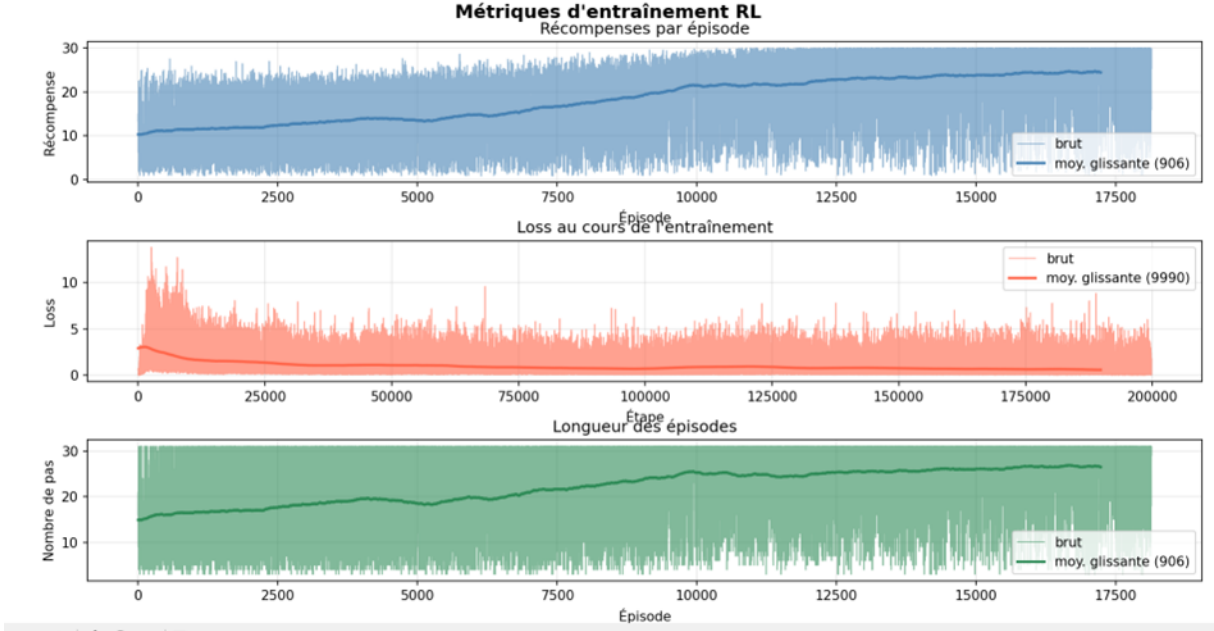


Figure 8: Training metrics for the Double DQN over 200 000 steps ($\approx 17\,800$ episodes). Top: per-episode reward (raw and 906-episode sliding mean). Middle: TD loss (raw and 9 990-step sliding mean). Bottom: episode length (raw and 906-episode sliding mean).

3.5 Quantitative evaluation results

I implemented the per-step speed distribution, reward distribution, seed stability, survival curve, and conditional action distribution visualizations used in the group evaluation.

Ablation study results (Figure 9). The five-model comparison over 3×50 evaluation episodes is reported in Figure 9.

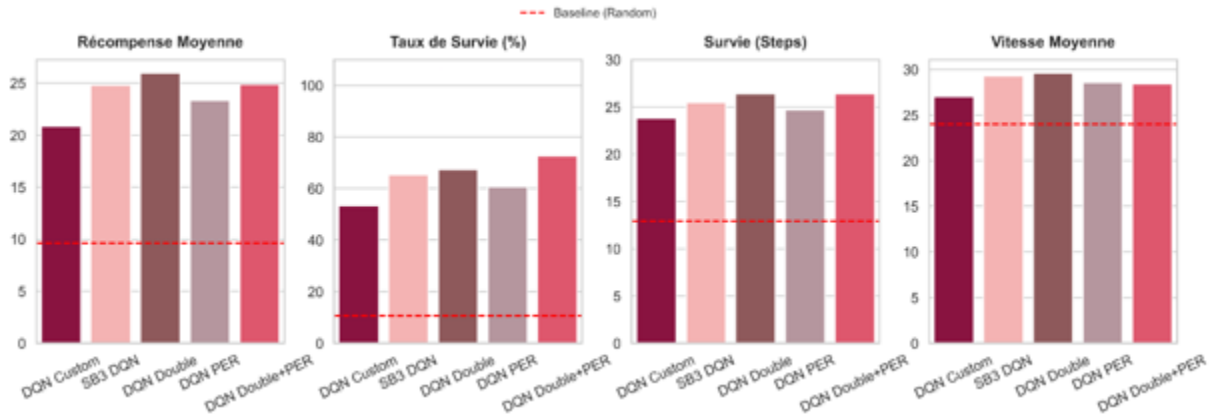


Figure 9: Full ablation study: five-model comparison over 3×50 evaluation episodes. *DQN Custom* is the Optuna fine-tuned vanilla DQN. Red dashed line: random baseline.

The **Double DQN** achieves the highest mean reward (≈ 26) and survival duration (≈ 27 steps). Relative to the fine-tuned DQN (≈ 21), this represents a $\approx 24\%$ improvement, placing the DDQN marginally above the SB3 baseline (≈ 25).

The most surprising result concerns **PER alone**, which yields only $\approx 61\%$ survival and ≈ 23 mean reward, underperforming both the DDQN and the SB3 baseline despite adding algorithmic complexity. A plausible explanation is that in *highway-v0*'s short-episode setting,

collision transitions receive systematically high TD-error and are therefore over-replayed, biasing the policy toward excessive caution and sacrificing speed reward. This over-prioritization effect is well documented in environments with sparse, high-magnitude penalties. The combination of **DDQN+PER** achieves the highest survival rate ($\approx 74\%$) while yielding a reward of ≈ 25 , suggesting that the DDQN’s reduced overestimation bias partially compensates for the PER over-prioritization, shifting the policy toward safety at a modest speed cost.

Fine-tuned DQN vs. Double DQN: answering the central question. The DDQN outperforms the fine-tuned DQN on all four metrics despite using *default, unoptimized hyperparameters*: mean reward improves by $\approx 24\%$ ($21 \rightarrow 26$), survival rate by 9 percentage points ($58\% \rightarrow 67\%$), and mean speed by ≈ 1 m/s. This finding directly answers the hypothesis posed in the abstract: **hyperparameter optimization alone does not suffice to close the gap with architectural improvements**. The maximization bias correction provided by the Double DQN yields independent gains beyond what Optuna-level tuning achieves on the vanilla architecture. Tuning the DDQN with the same Optuna pipeline would be required to fully disentangle both contributions; this is identified as a limitation in Section 4.

3.6 Behavioral analysis

I produced the two following figures by running the trained agents in evaluation mode ($\varepsilon = 0$) over 150 episodes and logging every step. State-conditional action distributions were then computed by binning the ego-vehicle’s normalized headway and speed from the kinematic observation matrix.

3.6.1 Action conditioned on headway to the nearest vehicle

Figure 10 reports the action distribution conditioned on the normalized relative distance to the nearest vehicle ahead, restricted to steps where a vehicle was detected. A left bin label indicates a close vehicle; a right bin label indicates a distant one.

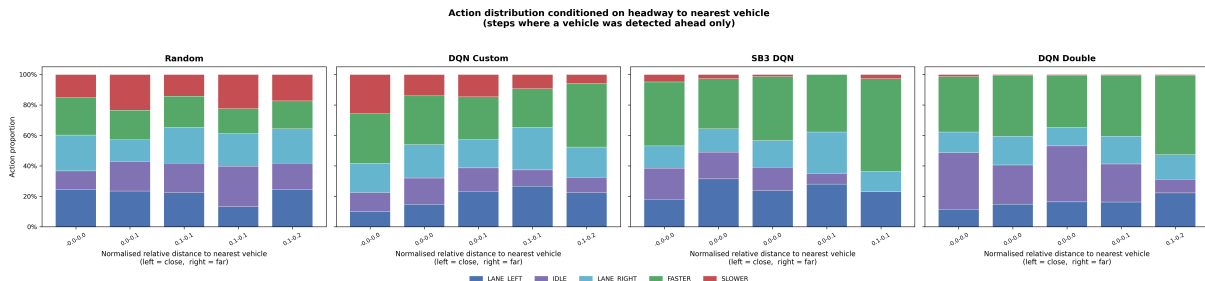


Figure 10: Action distribution conditioned on normalized headway to the nearest ahead vehicle (steps where a vehicle was detected only). Left bins correspond to close proximity; right bins to large headway. The four panels compare the random policy, the fine-tuned DQN, the SB3 DQN, and the Double DQN. (red = SLOWER, green = FASTER).

Random baseline. As expected, the random agent is headway-insensitive: the five actions are approximately equiprobable ($\approx 20\%$ each) across all proximity bins, with only minor statistical fluctuations. This provides a flat reference against which learned policies can be compared.

Fine-tuned DQN. The fine-tuned agent is clearly headway-sensitive. At close proximity, SLOWER accounts for $\approx 30\%$ of actions and IDLE for $\approx 25\%$, while FASTER is reduced to $\approx 20\%$. As the headway increases, this balance inverts sharply: FASTER rises to $\approx 60\%$ and SLOWER nearly vanishes ($\approx 5\%$). The agent has learned a *decelerate-when-close, accelerate-when-clear* strategy, which is mechanistically the correct response to the reward function: decelerating

avoids the -1.5 collision penalty while the $+0.7$ speed reward is maximized in free-running sections. Lane changes (LANE_LEFT and LANE_RIGHT) are present but moderate across all bins, consistent with the -0.02 lane-change penalty discouraging them.

SB3 DQN. The SB3 agent shows markedly reduced headway-sensitivity compared to the fine-tuned DQN. FASTER dominates at $\approx 60\text{--}65\%$ across all headway bins, including at close proximity. SLOWER is nearly absent ($\approx 2\text{--}3\%$) at all distances, and LANE_RIGHT is elevated ($\approx 15\%$). The SB3 policy can be characterized as *accelerate first, steer laterally if necessary*: it does not learn to decelerate as a primary collision-avoidance mechanism. This explains the SB3’s slightly lower survival rate ($\approx 65\%$) compared to the DDQN despite a competitive mean reward.

Double DQN. The DDQN exhibits a qualitatively different collision-avoidance strategy from both the fine-tuned DQN and SB3. At close proximity, LANE_LEFT dominates at $\approx 35\%$, substantially higher than in any other agent. FASTER is already elevated ($\approx 40\text{--}50\%$), and SLOWER is nearly absent ($< 5\%$) regardless of headway. As the vehicle ahead becomes more distant, FASTER grows further to $\approx 60\text{--}65\%$ and LANE_LEFT remains significant. The DDQN has therefore learned an *overtake-rather-than-brake* strategy: its primary response to proximity is a lane change to the left, rather than deceleration. This is consistent with reduced overestimation bias: by not inflating the Q-value of aggressive FASTER actions, the DDQN can more accurately evaluate the relative merit of a lateral avoidance manoeuvre. The near-total absence of SLOWER suggests that the DDQN essentially never decelerates; its safety margin comes from spatial reorganization rather than speed reduction.

3.6.2 Action conditioned on ego speed

Figure 11 reports the action distribution conditioned on the ego-vehicle’s speed bin. The speed bins are empirically determined from the visited states in evaluation rollouts and therefore differ across agents, reflecting each policy’s effective operating range.

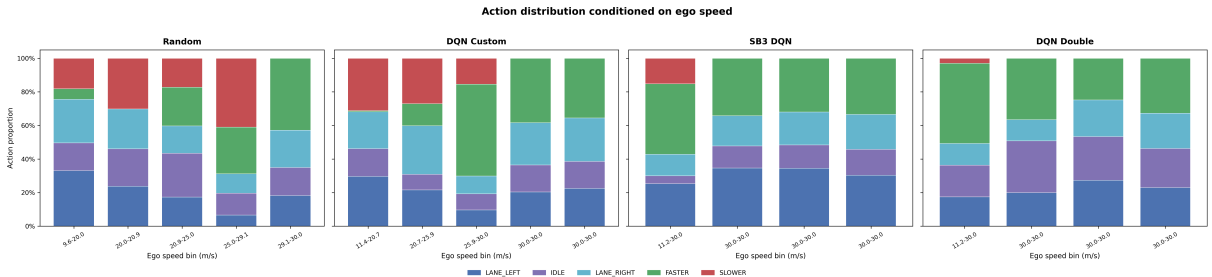


Figure 11: Action distribution conditioned on ego speed bin (m/s) during evaluation rollouts. Bins are empirically determined per agent. The four panels compare the random policy, the fine-tuned DQN, the SB3 DQN, and the Double DQN.

Random baseline. The random agent visits speed bins from 9.6 to 30 m/s and is mildly speed-sensitive by chance: FASTER slightly increases and SLOWER slightly decreases as speed grows, but the distribution remains broadly uniform.

Fine-tuned DQN. The fine-tuned agent’s speed bins span 11.4–30 m/s. At the lowest bin (11.4–20.7 m/s), SLOWER accounts for $\approx 25\%$ of actions, alongside FASTER at $\approx 35\%$ and IDLE at $\approx 20\%$. This diversity reflects the lower-speed regime where the agent is actively navigating around obstacles. A critical observation arises at the highest speed bin (30.0 m/s): FASTER still accounts for $\approx 60\%$ of actions even when the agent is already at the maximum speed attainable

in the environment. This *speed-saturation artefact* indicates that the fine-tuned policy has not learned that FASTER is a no-op at maximum speed: it continues to issue the command even when it cannot produce any further acceleration. This is a form of policy rigidity that may contribute to the late-episode crash observed in the survival curve (Section 2.2.3): an agent locked in a FASTER-dominant mode cannot decelerate rapidly enough when a vehicle cluster materializes.

SB3 DQN. The SB3 agent operates almost exclusively in the 30–30 m/s bin across all columns, indicating that the policy reaches maximum speed very early and stays there. FASTER is dominant at $\approx 75\text{--}80\%$ with almost no variation across bins. The SB3 policy is essentially a degenerate high-speed policy with a very narrow operating range.

Double DQN. Like the SB3 agent, the DDQN operates predominantly at maximum speed (most bins are 30–30 m/s), reflecting its aggressive acceleration strategy. However, the action composition differs fundamentally: LANE_LEFT accounts for $\approx 30\%$ of actions even at maximum speed, FASTER is at $\approx 60\%$, and SLOWER is negligible ($< 3\%$) across all bins. This confirms the finding from the headway analysis: the DDQN avoids obstacles through spatial reorganization (LANE_LEFT) rather than speed reduction. This strategy is more robust than the fine-tuned DQN’s deceleration approach *as long as a passable lane exists*, but fails catastrophically when the agent is already in the leftmost lane or all adjacent lanes are occupied. This constitutes the DDQN’s primary failure mode: a *blocked overtake*, wherein the agent’s dominant response (change left) is structurally unavailable.

3.6.3 Synthesis: three distinct collision-avoidance strategies

The behavioral analysis reveals that the three trained agents have converged to qualitatively distinct collision-avoidance strategies despite sharing the same reward function:

- The **fine-tuned DQN** uses a *decelerate-then-overtake* strategy: it reduces speed when a vehicle is close and accelerates when the lane is clear. Its failure mode is the speed-saturation artefact, which locks it in a FASTER-dominant mode at high speeds.
- The **SB3 DQN** uses a *fixed-speed lateral* strategy: it almost never decelerates and maintains maximum speed, relying on moderate lateral movement to avoid vehicles. Its reduced headway-sensitivity makes it prone to late-reaction collisions.
- The **Double DQN** uses an *overtake-first* strategy: its primary response at any speed and any headway is a left lane change, supplemented by sustained acceleration. This produces the highest mean reward and survival rate among the three, but generates a distinct failure mode when lateral manoeuvres are structurally unavailable.

These differences do not seem incidental: they are mechanistically linked to the respective training regimes. The fine-tuned DQN’s large replay buffer ($|\mathcal{B}| = 30\,000$) and low discount factor ($\gamma = 0.83$) encourage learning from a rich variety of transitions with short-horizon planning, which naturally produces a reactive deceleration policy. The DDQN’s reduced overestimation bias allows it to more accurately evaluate the long-term benefit of a lane change, which has a small immediate cost (-0.02) but a high expected future reward if it avoids a collision.

4 Conclusion

This report presented two personal contributions to the group project. First, I designed and executed a TPE-based Optuna hyperparameter search for the vanilla DQN. The key finding is that fine-tuning produces a qualitatively distinct policy: where the untuned model collapses

to a near-degenerate single-speed policy with a survival rate ($\approx 4\%$) indistinguishable from random, the fine-tuned model acquires active speed modulation and achieves $\approx 58\%$ survival. The fANOVA analysis identifies buffer capacity as the dominant hyperparameter (importance = 0.331) and reveals that a low discount factor ($\gamma \approx 0.83$) unexpectedly outperforms higher values.

Second, I implemented and analyzed a Double DQN, which outperforms the fine-tuned DQN on all four evaluation metrics despite using default, unoptimized hyperparameters (+24% mean reward, +9 percentage-point survival rate). The conditional action analysis shows that the performance gap is explained by a qualitatively different policy: the DDQN avoids obstacles through lane changes rather than deceleration, a strategy enabled by its reduced overestimation bias.

Limitations and future directions. The Optuna study was severely budget-constrained (10 trials, single seed). A controlled experiment applying the same Optuna pipeline to the DDQN would fully decouple the contributions of tuning and architecture. The behavioral analysis also suggests a more targeted extension: since the DDQN’s failure mode is a blocked overtake, training with a modified reward that penalizes leftmost-lane occupation would encourage a more diverse avoidance repertoire. More broadly, the observation window limitation (5 nearest vehicles) motivates richer representations such as attention-based architectures or graph neural networks capable of modeling inter-vehicle dependencies beyond the nearest-neighbor constraint.